

Project P5: Network Threat Detection & Monitoring

Sysmon Event ID 3 XML Parsing, Multi-Port Reconnaissance Triage, and Operational SOC Dashboarding

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TARGET MONITORED ENDPOINT Windows 11 (192.168.100.8)	ATTACKING SOURCE IP 192.168.100.6 (Kali Linux)	SIEM ARCHITECTURE Splunk Enterprise 10.4.3	INGESTION INDEX / TELEMETRY index=sysmon (Event ID 3)

1. Executive Summary

During operational threat hunting in the Splunk SOC laboratory, security alerts were generated indicating systematic network reconnaissance, high-volume connection velocity, and communication across unauthorized non-standard ports targeting the Windows 11 workstation (192.168.100.8). Telemetry collected via the Splunk Universal Forwarder from Microsoft Sysmon Event ID 3 (NetworkConnect) surfaced a three-phase adversary simulation:

Key Incident Findings: A total of **116 network connection events** were ingested and analyzed. Telemetry captured: (1) **Vertical port reconnaissance** from adversary host 192.168.100.6 probing listening endpoints on TCP 22 (OpenSSH) and TCP 3389 (RDP); (2) **26 suspicious connection events** directed to non-standard high port 8000 initiated by powershell.exe; and (3) an aggressive **connection velocity spike** reaching ~50 connections per 5-minute interval. All activities were correlated and flagged in real time.

2. Incident Scope & Environment Baseline

System Node	Network IP	Operating System	Role in Telemetry Pipeline
Target Endpoint	192.168.100.8	Windows 11 Enterprise (64-bit)	Monitored victim endpoint hosting Sysmon64 and Splunk Universal Forwarder 10.4.3.
Attacker Node	192.168.100.6	Kali Linux 2026 (Rolling)	Controlled adversary generating Nmap vertical scans and reconnaissance probes.
SIEM Server	192.168.100.7	Ubuntu Server 24.04 LTS	Splunk Enterprise 10.4.3 indexing server listening on TCP 9997 (Receiver) and Web 8000.
Gateway / Host	192.168.100.1	Windows Host Hypervisor	NAT network gateway routing management and administrative traffic.

3. Threat Scenarios & Adversary Behaviors Analyzed

- Scenario 1 – Vertical Port Scan (T1046 / T1595.002):** The adversary executed automated TCP service discovery probes targeting standard listening services (ports 22, 3389). Closed ports were silently dropped by Windows Defender Firewall, isolating process-bound connections.
- Scenario 2 – Suspicious / Backdoor High-Port Communication (T1571):** Telemetry identified process-level socket invocations over TCP 8000 originating from powershell.exe, representing C2 beaconing or staged exfiltration.
- Scenario 3 – Connection Velocity Burst & DoS Spike (T1498):** Rapid iterative socket connections were generated in short succession, exceeding the rate threshold of 20 connections per 5-minute window and driving total connection count to 116.

4. Telemetry Extraction & Search-Time XML Engineering

The Universal Forwarder on Windows 11 was configured with `renderXml = true`, transmitting Sysmon telemetry as structured XML. Because standard field extractions do not parse XML attributes by default, regular expression extraction rules were calibrated for the Windows Event XML schema:

```
index=sysmon earliest=-24h
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| search EventID=3
| rex field=_raw "<Data Name=['\"]SourceIp['\"]>(?!<src_ip>[^\"]+)</Data>"
| rex field=_raw "<Data Name=['\"]DestinationIp['\"]>(?!<dest_ip>[^\"]+)</Data>"
| rex field=_raw "<Data Name=['\"]DestinationPort['\"]>(?!<dest_port>\d+)</Data>"
| rex field=_raw "<Data Name=['\"]Image['\"]>(?!<process_image>[^\"]+)</Data>"
| table _time, src_ip, dest_ip, dest_port, process_image
| sort - _time
```

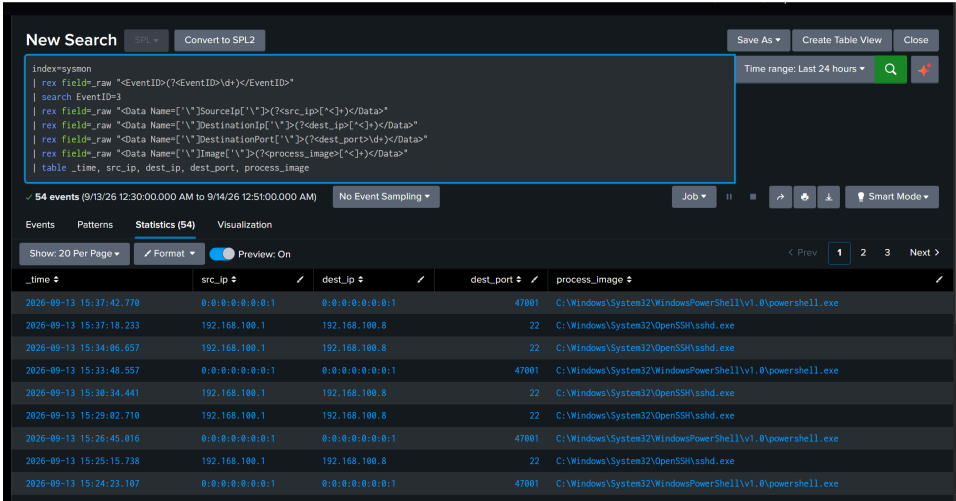


Figure 4.1: Splunk Search Head executing XML regular expression extraction for Sysmon Event ID 3 attributes.

5. Operational SOC Threat Hunting Dashboard

To provide SOC tier-1 and tier-2 analysts with immediate operational triage, an interactive Simple XML dark-theme dashboard was deployed. The dashboard monitors real-time connection volume, distinct port cardinality, uncommon port alerts, and live connection streams.

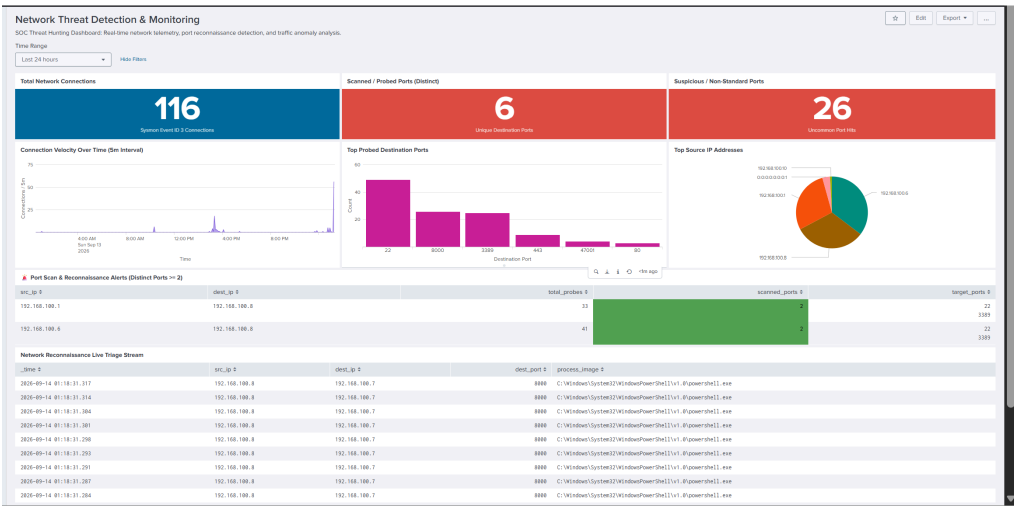


Figure 5.1: Operational dark-mode SOC dashboard displaying 116 connections, 6 distinct ports (Red Alert), 26 uncommon port hits (Red Alert), connection velocity burst spike, and live triage stream.

6. Port Scan Detection Search Execution

The production port scan detection query aggregates distinct destination ports per source IP. Probes from Kali Linux (192.168.100.6) targeting ports 22 and 3389 were flagged with 41 total connection attempts.

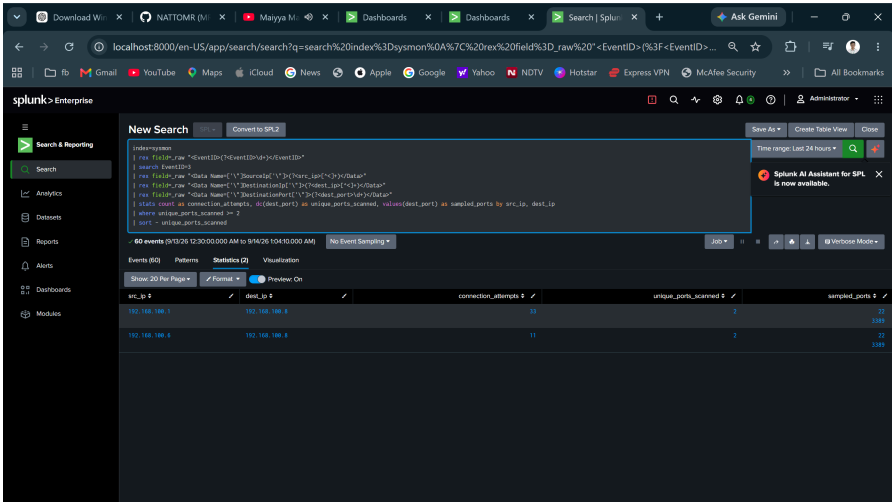


Figure 6.1: SPL detection query results isolating adversary IP 192.168.100.6 and gateway 192.168.100.1 with distinct destination port cardinality.

7. MITRE ATT&CK Framework Alignment

Tactic	Technique Name	ATT&CK ID	Lab Evidence & Detection Application
Reconnaissance	Active Scanning: Port Scanning	T1595.002	Systematic Nmap TCP connection probes from 192.168.100.6 targeting ports 22 and 3389.
Discovery	Network Service Discovery	T1046	Adversary enumeration to identify listening network services on Windows 11.
Command & Control	Non-Standard Port	T1571	Outbound high-volume burst over TCP 8000 originating from powershell.exe.
Execution	Command & Scripting: PowerShell	T1059.001	Automated programmatic socket invocations initiated through PowerShell scripts.
Impact	Network Denial of Service	T1498	Velocity flood spike testing endpoint connection rate limits and monitoring thresholds.

8. Analyst Recommendations & Remediation Playbook

1. **Inbound Firewall Hardening:** Configure Windows Defender Firewall to restrict inbound SSH (22) and RDP (3389) strictly to the trusted management IP (192.168.100.1), dropping unauthorized internal subnet probes:

Set-NetFirewallRule -Name "OpenSSH-Server-In-TCP" -RemoteAddress 192.168.100.1
Set-NetFirewallRule -DisplayGroup "Remote Desktop" -RemoteAddress 192.168.100.1
2. **Restricting Scripted Socket Connections:** Deploy AppLocker policies and enforce PowerShell Constrained Language Mode (CLM) to prevent unauthorized user-mode binaries from opening arbitrary outbound TCP sockets.
3. **Scheduled Splunk Real-Time Alert:** Operationalize port-scan-detection.spl as an alert scheduled every 5 minutes with a webhook to the SOC ticketing system whenever scanned_ports >= 2.
4. **Attacker Quarantine:** Add dynamic firewall drop rules at the hypervisor or switch layer to isolate 192.168.100.6 immediately upon alert generation.